

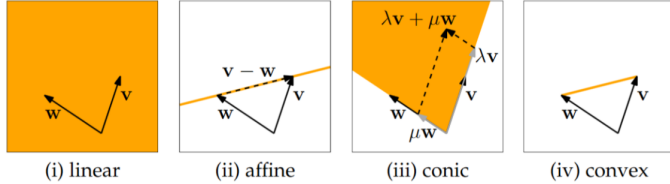
1 Vectors

Linear Combinations

D 1.4 If $v_1, \dots, v_n \in \mathbb{R}^m$ and $\lambda_1, \dots, \lambda_n \in \mathbb{R}$, then $\lambda_1 v_1 + \dots + \lambda_n v_n$ is a **linear combination** of $v_1, \dots, v_n$.

D 1.7 A linear combination $\lambda_1 v_1 + \dots + \lambda_n v_n$ is called

1. an **affine combination** if $\lambda_1 + \lambda_2 + \dots + \lambda_n = 1$.
2. a **conic combination** if $\lambda_i \geq 0$ for $i = 1, \dots, n$.
3. a **convex combination** if both affine and conic combination.



Scalar Product

D 1.9 Let $v, w \in \mathbb{R}^m$. Then the **scalar product** of v and w is the number $v \cdot w := v_1 w_1 + \dots + v_m w_m = \sum_{i=1}^m v_i w_i$

Cross product: $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$

O 1.10 Let $u, v, w \in \mathbb{R}^m$ and let $\lambda \in \mathbb{R}$. Then

1. $v \cdot w = w \cdot v$
2. $(\lambda v) \cdot w = \lambda(v \cdot w) = v \cdot \lambda w$
3. $u \cdot (v + w) = u \cdot v + u \cdot w$ and $(u + v) \cdot w = u \cdot w + v \cdot w$
4. $v \cdot v \geq 0$, with equality iff $v = 0$

D 1.11 Let $v \in \mathbb{R}^m$. The **Euclidian norm** of v is $\|v\| := \sqrt{v \cdot v}$

L 1.12 (**Cauchy-Schwarz inequality**) For any two vectors $v, w \in \mathbb{R}^m$:
 $|v \cdot w| \leq \|v\| \|w\|$

D 1.14 Let $v, w \in \mathbb{R}^m$ be two nonzero vectors. The angle between them is the unique α between 0 and π such that:

$$\cos(\alpha) = \frac{v \cdot w}{\|v\| \|w\|} \in [-1, 1]$$

Linear (in)dependence

D 1.16 Let $d \in \mathbb{R}^m, d \neq 0$. The set $H_d = \{v \in \mathbb{R}^m : v \cdot d = 0\}$ is called a **hyperplane through the origin**.

L 1.17 (**Triangle ineq.**) Let $v, w \in \mathbb{R}^m$. Then $\|v + w\| \leq \|v\| + \|w\|$.

D 1.21 Vectors $v_1, v_2, \dots, v_n$ are called **linearly dependent** if at least one of them is a linear combination of the others, i.e. if there exists an index $k \in [n]$ and scalars λ_j , such that $v_k = \sum_{j=1, j \neq k}^n \lambda_j v_j$. Otherwise,

they are called **linearly independent**.

L 1.22 Let $v_1, \dots, v_n \in \mathbb{R}^m$. The following statements are equivalent:

1. At least one of the vectors is a linear combination of the others.
2. There are scalars $\lambda_1, \dots, \lambda_n$ besides $0, \dots, 0$ such that $\sum_{j=1}^n \lambda_j v_j = \vec{0}$. We also say that $\vec{0}$ is a **nontrivial linear combination** of the vectors.

3. At least one of the vectors is a linear combination of the previous.

D 1.25 Let $v_1, \dots, v_n \in \mathbb{R}^m$. Their **span** is the set of all their linear combinations:

$$\text{Span}(v_1, \dots, v_n) := \left\{ \sum_{j=1}^n \lambda_j v_j \mid \lambda_j \in \mathbb{R} \text{ for all } j \in [n] \right\}$$

L 1.26 The span of a set of vectors does not change when adding a linear combination of its vectors to said set.

L 1.28 Let $v_1, \dots, v_m \in \mathbb{R}^m$ be linearly independent.
 $\text{Span}(v_1, \dots, v_m) = \mathbb{R}^m$

2 Matrices

D 2.3 Let $A = [a_{ij}]_{i=1}^m_{j=1}^m$ be an $m \times m$ square matrix. Then:

1. If $a_{ii} = 1$ for all i and $a_{ij} = 0$ for all $j \neq i$, then A is the **identity matrix**, denoted (in abuse of notation) by I in every dimension.
2. If $a_{ij} = 0$ for all $j \neq i$ (entries not on the diagonal are 0), then A is a **diagonal matrix**.
3. If $a_{ij} = 0$ for all $j < i$ (entries below the diagonal are 0), then A is an **upper triangular matrix**.
4. If $a_{ij} = 0$ for all $j > i$ (entries above the diagonal are 0), then A is a **lower triangular matrix**.
5. If $a_{ij} = a_{ji}$ for all i, j , then A is a **symmetric matrix**.

D 2.4 Let $A \in \mathbb{R}^{m \times n}$ be a matrix, $v_1, \dots, v_n$ its columns, and let $x \in \mathbb{R}^n$. Then the vector $Ax := \sum_{j=1}^n x_j v_j \in \mathbb{R}^m$ is called the **product** of A and x .

D 2.9 Let $A \in \mathbb{R}^{m \times n}$. The **column space** $C(A)$ of A is the span of the columns, $C(A) := \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$.

D 2.14 Let $A \in \mathbb{R}^{m \times n}$. The **row space** $R(A)$ of A is the column space of the transpose, $R(A) := C(A^T) \subseteq \mathbb{R}^n$.

D 2.17 Let $A \in \mathbb{R}^{m \times n}$. The **nullspace** of A is the set $N(A) = \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n$.

D 2.12 Let $A = [a_{ij}]_{i=1}^m_{j=1}^n$. The **transpose** of A is the $n \times m$ matrix $A^T := [a_{ji}]_{i=1}^n_{j=1}^m$.

L 2.40 Let $A \in \mathbb{R}^{a \times n}$ and $B \in \mathbb{R}^{n \times b}$. Then $(AB)^T = B^T A^T$.

Linear Transformations

D 2.21 A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m / T : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **linear transformation** / **linear functional** if the following linearity axiom holds for all $x_1, x_2 \in \mathbb{R}^n$ and all $\lambda_1, \lambda_2 \in \mathbb{R}$: $T(\lambda_1 x_1 + \lambda_2 x_2) = \lambda_1 T(x_1) + \lambda_2 T(x_2)$. In alternative, this can be proved with two axioms (for all $x, x' \in \mathbb{R}^n$ and all $\lambda \in \mathbb{R}$):

1. $T(x + x') = T(x) + T(x')$
2. $T(\lambda x) = \lambda T(x)$

D 2.27 Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. The following set is the **kernel** of T :

$$\text{Ker}(T) := \{x \in \mathbb{R}^n \mid T(x) = 0\} \subseteq \mathbb{R}^n$$

The following set is the **image** of T :

$$\text{Im}(T) := \{T(x) \mid x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$$

O 2.28/29 Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and $A \in \mathbb{R}^{m \times n}$ s.t. $T = T_A$. $\text{Im}(T) = C(A)$. $\text{Ker}(T) = N(A)$.

CR decomposition

T 2.46 Let $A \in \mathbb{R}^{m \times n}$ of rank r . Let C be the $m \times r$ submatrix of A containing the independent columns. Then there exists a unique $r \times n$ matrix R such that $A = CR^T$. (This **CR decomposition** can be constructed using Gauss-Jordan elimination.) The columns of R^T contain the scalars that we need in order to write each column as a scalar multiple of the first one.

Inverse Matrices

D 2.48 Let X, Y be sets and $f : X \rightarrow Y$ a function.

1. f is called **injective** if for every $y \in Y$, there is at most one $x \in X$ with $f(x) = y$.
2. f is called **surjective** if for every $y \in Y$, there is at least one $x \in X$ with $f(x) = y$.
3. f is called **bijective** (undoable) if f is both injective and surjective.
4. The inverse of a bijective function f is: $f^{-1} : Y \rightarrow X, y \mapsto \text{the unique } x \in X \text{ such that } f(x) = y$.

L 2.53 Let $A \in \mathbb{R}^{m \times m}$. The following statements are equivalent:

1. $T_A : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is bijective.
2. There is $B \in \mathbb{R}^{m \times m}$ such that $BA = I$.
3. The columns of A are linearly independent.

D 2.55/57 Let $M \in \mathbb{R}^{m \times m}$. M is called **invertible** if there exists a matrix $M^{-1} \in \mathbb{R}^{m \times m}$ (called the **inverse** of M) such that $MM^{-1} = M^{-1}M = I$.

L 2.59 Let $A, B \in \mathbb{R}^{m \times m}$ be invertible matrices. Then AB is also invertible, and $(AB)^{-1} = B^{-1}A^{-1}$.

L 2.60 Let $A \in \mathbb{R}^{m \times m}$ be an invertible matrix. Then A^T is also invertible, and $(A^T)^{-1} = (A^{-1})^T$.

4 The Three Fundamental Subspaces

Vector Spaces

D 4.1 A **vector space** is a triple $(V, +, \cdot)$ where V is a set, $+: V \times V \rightarrow V$ is a function (vector addition), and $\cdot : \mathbb{R} \times V \rightarrow V$ is a function (scalar multiplication), satisfying the following axioms for all $u, v, w \in V$ and all $\lambda, \mu \in \mathbb{R}$:

1. $v + w = w + v$ (**commutativity**)
2. $u + (v + w) = (u + v) + w$ (**associativity**)
3. There is a vector 0 such that $v + 0 = v$. (**zero vector**)
4. There is a vector $-v$ such that $v + (-v) = 0$. (**negative vector**)
5. $1 \cdot v = v$ (**identity element**)
6. $(\lambda \cdot \mu)v = \lambda \cdot (\mu \cdot v)$ (**compatibility of · and · in ℝ**)
7. $\lambda(v + w) = \lambda v + \lambda w$ (**distributivity over +**)
8. $(\lambda + \mu)v = \lambda v + \mu v$ (**distributivity over + in ℝ**)

T 4.4 Let $\mathbb{R}[x]$ be the set of polynomials in one variable x . Given polynomials $p = \sum_{i=0}^m p_i x^i$ and $q = \sum_{i=0}^n q_i x^i$, and a scalar $\lambda \in \mathbb{R}$ we define:

$$p + q = \sum_{i=0}^{\max(m,n)} (p_i + q_i) x^i \quad \text{and} \quad \lambda p = \sum_{i=0}^m (\lambda p_i) x^i$$

We set $p_i = 0$ for $i > m$ and $q_i = 0$ for $i > n$. Then $(\mathbb{R}[x], +, \cdot)$ is a vector space.

D 4.8 Let V be a vector space. A nonempty subset $U \subseteq V$ is called a **subspace** of V if the following two axioms of a subspace are true for all $v, w \in U$ and all $\lambda \in \mathbb{R}$:

1. $v + w \in U$
2. $\lambda v \in U$

L 4.14 Let $U \subseteq V$ be a subspace of a vector space V . Then U is also a vector space (with the same “+” and “ $\cdot$ ” as V).

L 4.16 Let V be a vector space. Every linear combination of V is again in V .

Bases and Dimension

D 4.17 Let V be a vectorspace, and $G \subseteq V$ a (possibly infinite) subset of vectors. Then $\text{Span}(G)$ is the set of all linear combinations of G . The set G is called linearly independent if no vector $v \in G$ is a linear combination of $G \setminus \{v\}$.

D 4.18 Let V be a vector space. A subset $B \subseteq V$ of vectors is called **basis** of V if B is linearly independent and $\text{Span}(B) = V$.

L 4.19 Let $A \in \mathbb{R}^{m \times n}$. The set of independent columns of A is a basis of the column space $\mathbf{C}(A)$.

L 4.23 (**Steinitz exchange L**) Let V be a finitely generated vector space, $F \subseteq V$ a finite set of linearly independent vectors, and $G \subseteq V$ a finite set of vectors with $\text{Span}(G) = V$. Then the following two statements hold:

1. $|F| \leq |G|$
2. There exists a subset $E \subseteq G$ of size $|G| - |F|$ such that $\text{Span}(F \cup E) = V$.

D 4.21 A vector space V is called **finitely generated** if there exists a finite subset $G \subseteq V$ with $\text{Span}(G) = V$.

T 4.22 Let V be a finitely generated vector space, and let $G \subseteq V$ be a finite subset with $\text{Span}(G) = V$. Then V has a basis $B \subseteq G$.

T 4.24 Let V be a finitely generated vector space and $B, B' \subseteq V$ two bases of V . Then $|B| = |B'|$ (all bases have the same size).

D 4.25 Let V be a finitely generated vector space. Then $\dim(V)$, the **dimension** of V , is the size of any basis B of V .

T 4.31 Let $A \in \mathbb{R}^{m \times n}$ and let R in RREF($j_1, \dots, j_r$) be the result of Gauss-Jordan elimination on A . Then A has independent columns at indices $j_1, \dots, j_r$ and these form a basis of the column space $\mathbf{C}(A)$. Hence $\dim(\mathbf{C}(A)) = r = \text{rank}(A)$.

L Let $A \in \mathbb{R}^{m \times n}$ and $M \in \mathbb{R}^{m \times m}$ and invertible. Then $\mathbf{R}(A) = \mathbf{R}(MA)$.

T 4.32 Let $A \in \mathbb{R}^{m \times n}$ and let R_0 in RREF($j_1, \dots, j_r$) be the result of Gauss-Jordan elimination on A . Then the first r rows of R_0 form a basis of $\mathbf{R}(A)$, and $\dim(\mathbf{R}(A)) = r = \text{rank}(A)$.

T 4.33 Let $A \in \mathbb{R}^{m \times n}$. Then $\text{rank}(A) = \text{rank}(A^\top)$.

L Let $A \in \mathbb{R}^{m \times n}$ and $M \in \mathbb{R}^{m \times m}$ and invertible. Then A and MA have the same nullspace $\mathbf{N}(A) = \mathbf{N}(MA)$.

T 4.36 Let $A \in \mathbb{R}^{m \times n}$, and let R in RREF($j_1, \dots, j_r$) be the result of Gauss-Jordan elimination on A . Let Q be the $r \times (n - r)$ submatrix of R consisting of the first r rows and columns $k_1, \dots, k_{n-r}$. A basis of $\mathbf{N}(A)$ is given by the $n - r$ vectors $v_1, \dots, v_{n-r}$. Then $\dim(\mathbf{N}(A)) = n - r = n - \text{rank}(A)$.

D 4.37 Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. The **solution space** of $Ax = b$ is the set of all solutions x : $\text{Sol}(A, b) := \{x \in \mathbb{R}^n \mid Ax = b\} \subseteq \mathbb{R}^n$.

If $b \neq 0$, however, the solution space is not a subspace of $\mathbb{R}^n$, as it doesn't contain the zero vector.

T 4.38 Let $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$. Let s be some solution of $Ax = b$. Then $\text{Sol}(A, b) = \{s + x \mid x \in \mathbf{N}(A)\}$.

Hence, we can compute $\text{Sol}(A, b)$, despite it not being a subspace. To describe all solutions, we just need *some* solution and a basis of $\mathbf{N}(A)$.

T 4.39 Let $A \in \mathbb{R}^{m \times n}$ of rank r . If $Ax = b$ has a solution, then $\text{Sol}(A, b)$ has dimension $n - r$, where $\dim(\text{Sol}(A, b)) := \dim(\mathbf{N}(A))$.

T 4.40 Let $A \in \mathbb{R}^{m \times n}$ of rank m . Then $Ax = b$ has a solution for every $b \in \mathbb{R}^m$.

5 Orthogonality

D 5.1.1 Two vectors $v, w \in \mathbb{R}^n$ are called **orthogonal** if $v^\top w = 0$. Two subspaces V and W are orthogonal if for all $v \in V$ and all $w \in W$, the vectors v and w are orthogonal.

L 5.1.2 Let $v_1, \dots, v_k$ be a basis of subspace V . Let $w_1, \dots, w_l$ be a basis of subspace W . V and W are orthogonal iff v_i and w_j are orthogonal for all $i \in \{1, \dots, k\}$ and $j \in \{1, \dots, l\}$.

L 5.1.3 Let V and W be two orthogonal subspaces of $\mathbb{R}^n$. Let $v_1, \dots, v_k$ be a basis of subspace V . Let $w_1, \dots, w_l$ be a basis of subspace W . The set of vectors $\{v_1, \dots, v_k, w_1, \dots, w_l\}$ is linearly independent.

C 5.1.4 Let V and W be orthogonal subspaces. Then $V \cap W = \{0\}$. Moreover, if $\dim(V) = k$ and $\dim(W) = l$, then $\dim(\{\lambda v + \mu w \mid \lambda, \mu \in \mathbb{R}, v \in V, w \in W\}) = k + l \leq n$.

D 5.1.5 Let V be a subspace of $\mathbb{R}^n$. We define the **orthogonal complement** of V as: $V^\perp = \{w \in \mathbb{R}^n \mid w^\top v = 0 \text{ for all } v \in V\}$.

T 5.1.6 Let $A \in \mathbb{R}^{m \times n}$, then $\mathbf{N}(A) = \mathbf{C}(A^\top)^\perp = \mathbf{R}(A)^\perp$.

T 5.1.7 Let V, W be orthogonal subspaces of $\mathbb{R}^n$. The following statements are equivalent:

1. $W = V^\perp$
2. $\dim(V) + \dim(W) = n$
3. Every $u \in \mathbb{R}^n$ can be written as $u = v + w$ with unique vectors $v \in V, w \in W$.

L 5.1.10 Let $A \in \mathbb{R}^{m \times n}$. Then $\mathbf{N}(A) = \mathbf{N}(A^\top A)$ and $\mathbf{C}(A^\top) = \mathbf{C}(A^\top A)$.

Projections

D 5.2.1 The **projection** of a vector $b \in \mathbb{R}^m$ on a subspace S (of $\mathbb{R}^m$) is the point in S that is closest to b . In other words: $\text{proj}_S(b) = \arg\min_{p \in S} \|b - p\|$.

L 5.2.2 Let $a \in \mathbb{R}^m \setminus \{0\}$ and $S = \{\lambda a \mid \lambda \in \mathbb{R}\}$. $\text{proj}_S(b) = \frac{a a^\top}{a^\top a} b$.

L 5.2.3 The projection of a vector $b \in \mathbb{R}^m$ to the subspace $S = \mathbf{C}(A)$ can be written as $\text{proj}_S(b) = A\hat{x}$, whereby $\hat{x} = \min_{x \in \mathbb{R}^n} \|Ax - b\|^2$ satisfies $A^\top A\hat{x} = A^\top b$.

Note that, given the QR decomposition of A , $\hat{x}$ satisfies $R\hat{x} = Q^\top b$ and $\hat{x} = R^{-1}Q^\top b$ if A has independent columns (and thus R is invertible)

L 5.2.4 $A^\top A$ is invertible iff A has linearly independent columns.

T 5.2.5 Let S be a subspace of $\mathbb{R}^m$ and A a matrix whose columns are a basis of S . The projection of $b \in \mathbb{R}^m$ to S is given by $\text{proj}_S(b) = Pb$, where $P = A(A^\top A)^{-1}A^\top$ is the **projection matrix**.

6 Applications of orth. and proj.

Set of all solutions to a set of linear equations

L 6.2.1 Let $A \in \mathbb{R}^{m \times n}$. Let $x, y \in \mathbf{C}(A^\top)$. We have that $Ax = Ay \iff x = y$.

T 6.2.2 Suppose that $\{x \in \mathbb{R}^n \mid Ax = b\} \neq \emptyset$. Then $\{x \in \mathbb{R}^n \mid Ax = b\} = x_1 + \mathbf{N}(A)$ where $x_1 \in \mathbf{R}(A)$ is unique such that $Ax_1 = b$.

C 6.2.3 Suppose that $\{x \in \mathbb{R}^n \mid Ax = b\} \neq \emptyset$. Then there exists a unique vector $x_1 \in \mathbf{C}(A^\top A)$ such that $Ax_1 = b$.

T 6.2.4 $\{x \in \mathbb{R}^n \mid Ax = b\} = \emptyset \iff \{z \in \mathbb{R}^m \mid A^\top z = 0, b^\top z = 1\} \neq \emptyset$.

Orthonormal Vectors and Orthogonal Matrices

D 6.3.1 Vectors $q_1, \dots, q_n \in \mathbb{R}^m$ are **orthonormal** if they are orthogonal and have norm 1. In other words, for all $i, j \in \{1, \dots, n\}$, we have $q_i^\top q_j = \delta_{ij}$, where δ_{ij} is the Kronecker delta ($\delta_{ij} = \{0 \text{ if } i \neq j, 1 \text{ if } i = j\}$).

D 6.3.3 A square matrix $Q \in \mathbb{R}^{n \times n}$ is an **orthogonal matrix** when $Q^\top Q = I$. In this case, $QQ^\top = I$, $Q^{-1} = Q^\top$, and the columns of Q form an orthonormal basis for $\mathbb{R}^n$.

P 6.3.6 Orthogonal matrices preserve norm and inner product of vectors. In other words, if $Q \in \mathbb{R}^{n \times n}$ is orthogonal, then, for all $x, y \in \mathbb{R}^n$, we have: $\|Qx\| = \|x\|$ and $(Qx)^\top (Qy) = x^\top y$.

D 6.3.10 Let $A \in \mathbb{R}^{m \times n}$ with linearly independent columns. The **QR decomposition** is given by $A = QR$, where Q is an $m \times n$ matrix with orthonormal columns (they are the output of the Gram-Schmidt algorithm on the columns of A) and R is an upper triangular matrix given by $R = Q^\top A$ (and is invertible if A has independent columns).

Pseudoinverses (Moore-Penrose Inverse)

D 6.4.1 For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = n$, we define the **pseudoinverse (full column rank)** $A^\dagger \in \mathbb{R}^{n \times m}$ of A as $A^\dagger = (A^\top A)^{-1}A^\top$ ($A^\dagger A = I$).

D 6.4.3 For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$, we define the **pseudoinverse (full row rank)** $A^\dagger \in \mathbb{R}^{n \times m}$ of A as $A^\dagger = A^\top (AA^\top)^{-1}$ ($AA^\dagger = I$).

P 6.4.6 For a full row rank matrix A , the (unique) solution to $\min_{x \in \mathbb{R}^n} \|x\|^2$ such that $Ax = b$ is given by the vector $\hat{x} = A^\dagger b$.

D 6.4.7 For $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = r$ and CR decomposition $A = CR$, we define the **pseudoinverse** $A^\dagger$ as $A^\dagger = R^\dagger C^\dagger$.

L 6.4.8 Given $A \in \mathbb{R}^{m \times n}$ and a vector $b \in \mathbb{R}^n$, the (unique) solution to $\min_{x \in \mathbb{R}^n} \|x\|^2$ such that $A^\top Ax = A^\top b$ is given by $\hat{x} = A^\dagger b$.

T 6.4.10 Let $A \in \mathbb{R}^{m \times n}$. Then:

1. $AA^\dagger A = A$
2. $A^\dagger AA^\dagger = A^\dagger$

3. $AA^\dagger$ is symmetric. It is the proj. matrix for projection on $\mathbf{C}(A)$.
4. $A^\dagger A$ is symmetric. It is the proj. matrix for projection on $\mathbf{C}(A^\top)$.
5. $(A^\top)^\dagger = (A^\dagger)^\top$

7 The Determinant

D 7.2.1 Given a *permutation* $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ of n elements, its *sign* $\text{sgn}(\sigma)$ can be 1 or -1 . Then sign counts the parity of the number of pairs of elements that are out of order (sometimes called inversions) after applying the permutation. In other words, for all pairs $(i, j) \in \{1, \dots, n\} \times \{1, \dots, n\}$:

$$\text{sgn}(\sigma) = \begin{cases} 1 & \text{if } |\{(i, j) \mid i < j, \sigma(i) > \sigma(j)\}| \text{ even} \\ -1 & \text{if } |\{(i, j) \mid i < j, \sigma(i) > \sigma(j)\}| \text{ odd} \end{cases}$$

D 7.2.3 Let $A \in \mathbb{R}^{n \times n}$ be a square matrix and let Π_n be the set of all permutations of n elements. Then the *determinant* $\det(A)$ is defined as:

$$\det(A) = \sum_{\sigma \in \Pi_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i, \sigma(i)}$$

P 7.2.4

1. Given a permutation matrix $P \in \mathbb{R}^{n \times n}$ corresponding to a permutation σ , then $\det(P) = \text{sgn}(\sigma)$. We sometimes also write $\text{sgn}(P)$.
2. Given a triangular (either upper or lower) matrix $T \in \mathbb{R}^{n \times n}$, we have $\det(T) = \prod_{k=1}^n T_{kk}$.
3. If $Q \in \mathbb{R}^{n \times n}$ is an orthogonal matrix, then $\det(Q) = 1$ or $\det(Q) = -1$.

T 7.2.5 Given a matrix $A \in \mathbb{R}^{n \times n}$, we have $\det(A^\top) = \det(A)$.

T 7.2.6

1. $A \in \mathbb{R}^{n \times n}$ is invertible iff $\det(A) \neq 0$.
2. Given $A, B \in \mathbb{R}^{n \times n}$, we have $\det(AB) = \det(A)\det(B)$.
3. Given $A \in \mathbb{R}^{n \times n}$ such that $\det(A) \neq 0$, then A is invertible, and $\det(A^{-1}) = \frac{1}{\det(A)}$.

D 7.3.1 Given $A \in \mathbb{R}^{n \times n}$, for each $1 \leq i, j \leq n$ let $\mathcal{A}_{ij}$ denote the $(n-1) \times (n-1)$ matrix obtained by removing row i and column j from A . Then we define the *co-factors of A* as $C_{ij} = (-1)^{i+j} \det(\mathcal{A}_{ij})$.

P 7.3.2 Let $A \in \mathbb{R}^{n \times n}$, then for any $1 \leq i \leq n$: $\det(A) = \sum_{j=1}^n A_{ij} C_{ij}$.

P 7.3.3 Let $A \in \mathbb{R}^{n \times n}$ with $\det(A) \neq 0$ and let C be the $n \times n$ matrix with the co-factors of A as entries. Then $A^{-1} = \frac{1}{\det(A)} C^\top$.

P 7.3.5 (*Cramer's rule*) Let $A \in \mathbb{R}^{n \times n}$ such that $\det(A) \neq 0$ and $b \in \mathbb{R}^n$. Moreover, let $\mathcal{B}_j$ be the matrix obtained from A by replacing the j -th column of A with the vector b . Then the solution $x \in \mathbb{R}^n$ of $Ax = b$ is given by $x_j = \frac{\det(\mathcal{B}_j)}{\det(A)}$.

8 Eigenvalues and Eigenvectors

Complex Numbers

The *complex numbers* are of the form $z = a + ib$ for $a \in \mathbb{R}$ and $b \in \mathbb{R}$: $\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$. Hereby i is defined as $i^2 = -1$.

Given $z \in \mathbb{C}$ with $z = a + ib$, we have the following notation:

1. $\Re(a + ib) := a$ (*real part* of z)
2. $\Im(a + ib) := b$ (*imaginary part* of z)
3. $|z| := \sqrt{a^2 + b^2}$ (*modulus* of z)
4. $a + ib := a - ib$ (*complex conjugate* of z)
5. $A^H = A^* := \overline{A}^\top$ (*conjugate/hermitian transpose* of A)
6. $\|v\|^2 = v^* v = \overline{v}^\top v$ (*modulus* of vn)
7. $\langle v, w \rangle = w^* v$ (*inner/dot product* in $\mathbb{C}^n$)

We also have $|z|^2 = z\overline{z}$, $z_1 z_2 = z_2 z_1$, $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$, $\frac{1}{\overline{z}} = \frac{\overline{z}}{|z|^2}$.

The natural operation of "transposing" for complex vectors and matrices is that of the "*conjugate transpose*" or "hermitian transpose", denoted by A^* , or sometimes A^H : $v^* = \overline{v}^\top$ and $A^* = \overline{A}^\top$.

The *inner product* (or dot-product) in $\mathbb{C}^n$ is given by $\langle v, w \rangle = w^* v$.

T 8.1.2 (*Fundamental T of Algebra*) Any degree n non-constant ($n \geq 1$) polynomial $P(z)$ has a zero (root) $\lambda \in \mathbb{C}$, such that $P(\lambda) = 0$.

C 8.1.3 Any degree n non-constant ($n \geq 1$) polynomial $P(z) = \alpha_n z^n + \dots + \alpha_1 z + \alpha_0$ has n zeros (roots) $\lambda_1, \dots, \lambda_n \in \mathbb{C}$, perhaps with repetitions, such that $P(z) = \alpha_n (z - \lambda_1) \dots (z - \lambda_n)$.

The number of times $\lambda \in \mathbb{C}$ appears in this expansion is called the *algebraic multiplicity* of the zero.

Eigenvalues and Eigenvectors

D 8.2.1 Given $A \in \mathbb{R}^{n \times n}$, we say $\lambda \in \mathbb{C}$ is an *eigenvalue* of A and $v \in \mathbb{C}^n \setminus \{0\}$ is an *eigenvector* of A , associated with the eigenvalue λ , when the following holds: $Av = \lambda v$.

We call them an eigenvalue-eigenvector pair. If $\lambda \in \mathbb{R}$, then we call λ a *real* eigenvalue, and the associated eigenvalue-eigenvector pair a *real* eigenvalue-eigenvector pair.

L 8.2.3 Let $A \in \mathbb{R}^{n \times n}$. $\lambda \in \mathbb{R}$ is a (real) eigenvalue of A iff $\det(A - \lambda I) = 0$. A vector $v \in \mathbb{R}^n \setminus \{0\}$ is an eigenvector associated with the eigenvalue λ iff it is a non-zero element of $\mathbf{N}(A - \lambda I)$.

P 8.2.7 Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix. If $\lambda \in \mathbb{C}$ is an eigenvalue of Q , the $|\lambda| = 1$.

L 8.2.8 Let $A \in \mathbb{R}^{n \times n}$. If (λ, v) is an eigenvalue-eigenvector pair, so is $(\overline{\lambda}, \overline{v})$.

P 8.3.1

1. If λ and v are an eigenvalue-eigenvector pair of matrix A , then, for $k \geq 1$, λ^k and v are an eigenvalue-eigenvector pair of the matrix A^k .
2. Let A be an invertible matrix. If λ and v are an eigenvalue-eigenvector pair of the matrix A , then $\frac{1}{\lambda}$ and v are an eigenvalue-eigenvector pair of the matrix A^{-1} .

L 8.3.2 Let $A \in \mathbb{R}^{n \times n}$ and let $v_1, \dots, v_k \in \mathbb{R}^n$ be eigenvectors corresponding to eigenvalues $\lambda_1, \dots, \lambda_k \in \mathbb{R}$. If $\lambda_1, \dots, \lambda_k$ are all distinct, the eigenvectors $v_1, \dots, v_k$ are linearly independent.

T 8.3.3 Let $A \in \mathbb{R}^{n \times n}$ with n distinct real eigenvalues (the n zeros of $\det(A - \lambda I)$ are all distinct). Then there is a basis of $\mathbb{R}^n$, $v_1, \dots, v_n$, made up of eigenvectors of A .

D 8.3.4 (*Characteristic polynomial*) Let $A \in \mathbb{R}^{n \times n}$. Then $\det(zI - A)$ is a polynomial, in λ , of degree n , which can according to C 8.1.3 be factorised as follows: $\det(zI - A) = (z - \lambda_1) \dots (z - \lambda_n)$.

This polynomial is called the characteristic polynomial of A , and its roots $\lambda_1, \dots, \lambda_n \in \mathbb{C}$ are the eigenvalues of A . The number of times an eigenvalue shows up is called *algebraic multiplicity* of that eigenvalue.

The *trace* of A is defined as $\text{Tr}(A) = \sum_{i=1}^n A_{ii}$.

P 8.3.5 Given $A \in \mathbb{R}^{n \times n}$ the eigenvalues of A are the same as the ones of $A^\top$.

L 8.3.6 Let $A \in \mathbb{R}^{n \times n}$ and $\lambda_1, \dots, \lambda_n$ its n eigenvalues as they show up as zeroes in its characteristic polynomial, then:

1. $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$
2. $\det(A) = \prod_{i=1}^n \lambda_i$

L 8.3.7 For matrices $A, B, C \in \mathbb{R}^{n \times n}$, we have: $\text{Tr}(AB) = \text{Tr}(BA)$, $\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)$.

9 Diagonalisation

T 9.1.1 Let $A \in \mathbb{R}^{n \times n}$ be a matrix with a complete set of real eigenvectors and let $v_1, \dots, v_n \in \mathbb{R}^n$ be a basis formed with eigenvectors of A and let $\lambda_1, \dots, \lambda_n$ be the associated eigenvalues (λ_i associated to v_i). Let V be the matrix whose columns are the eigenvectors. Moreover, let Λ be a diagonal matrix with $\Lambda_{ii} = \lambda_i$ and $\Lambda_{ij} = 0$ for all $i \neq j$. Then $A = V\Lambda V^{-1}$.

D 9.1.2 $A \in \mathbb{R}^{n \times n}$ is called a *diagonalizable matrix* if there exists an inverse matrix V such that $V^{-1}AV = \Lambda$, where Λ is a diagonal matrix. $A = V\Lambda V^{-1}$.

D 9.1.3 If, given a matrix $A \in \mathbb{R}^{n \times n}$, we can build a basis of $\mathbb{R}^n$ with eigenvectors of A , we say that A has a *complete set of eigenvectors*.

P 9.1.6 Let P be a projection matrix on the subspace $U \subseteq \mathbb{R}^n$. Then P has two eigenvalues, 0 and 1, and a complete set of real eigenvectors.

D 9.1.7 We say that $A \in \mathbb{R}^{n \times n}$ and $B \in \mathbb{R}^{n \times n}$ are *similar matrices* if there exists an invertible matrix S such that $B = S^{-1}AS$.

P 9.1.8 Similar matrices have the same eigenvalues.

D 9.1.10 Given a matrix $A \in \mathbb{R}^{n \times n}$ and an eigenvalue λ of A , we call the dimension of $\mathbf{N}(A - \lambda I)$ the *geometric multiplicity* of λ .

L 9.1.11 A matrix has a complete set of real eigenvectors if all its eigenvalues are real and the geometric multiplicities are the same as the algebraic multiplicities of all its eigenvalues.

Spectral Theorem

T 9.2.1 (*Spectral Theorem*) Any symmetric matrix $A \in \mathbb{R}^{n \times n}$ has n real eigenvalues and an orthonormal basis of $\mathbb{R}^n$ made of eigenvectors of A .

C 9.2.2 Let $A \in \mathbb{R}^{n \times n}$ be a symmetric matrix and let $\Lambda \in \mathbb{R}^{n \times n}$ be a diagonal matrix with the eigenvalues of A on its diagonal. Then there exists an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ (whose columns are eigenvectors of A) such that $A = V\Lambda V^\top$.

C 9.2.4 The rank of a real symmetric matrix is the number of non-zero eigenvalues (counting repetitions).

P 9.2.6 $A \in \mathbb{R}^{n \times n}$, symmetric, $v_1, \dots, v_n$ orthonormal basis of eigenvectors of A and $\lambda_1, \dots, \lambda_n$ the associated eigenvalues. Then $A = \sum_{k=1}^n \lambda_k v_k v_k^\top$.

L 9.2.7 $A \in \mathbb{R}^{n \times n}$ symmetric, $\lambda_1 \neq \lambda_2 \in \mathbb{R}$ two eigenvalues of A with corresponding eigenvectors v_1, v_2 . Then, v_1 and v_2 are orthogonal.

L 9.2.8 $A \in \mathbb{R}^{n \times n}$ symmetric, $\lambda \in \mathbb{C}$ eigenvalue of A . Then, $\lambda \in \mathbb{R}$.

P 9.2.10 Given a symmetric matrix $A \in \mathbb{R}^{n \times n}$ the *Rayleigh quotient*, defined for $x \in \mathbb{R}^n \setminus \{0\}$, as $R(x) = \frac{x^\top A x}{x^\top x}$ attains its maximum at $R(v_{\max}) = \lambda_{\max}$ and its minimum at $R(v_{\min}) = \lambda_{\min}$, where $\lambda_{\max}$ and $\lambda_{\min}$ are the largest and smallest eigenvalue of A , respectively, and $v_{\max}$ and $v_{\min}$ their associated eigenvectors.

D 9.2.11 A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is said to be *positive semidefinite* (PSD) if all its eigenvalues are non-negative. If all its eigenvalues are, on the other hand, strictly positive, then we say A is *positive definite* (PD).

P 9.2.12 A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is PSD iff $x^\top A x \geq 0$ for all $x \in \mathbb{R}^n$. Analogously, a symmetric matrix $A \in \mathbb{R}^{n \times n}$ is PD iff $x^\top A x > 0$ for all $x \in \mathbb{R}^n \setminus \{0\}$.

D 9.2.13 Given n vectors, $v_1, \dots, v_n \in \mathbb{R}^m$, we call their *Gram matrix* the $n \times n$ matrix of their inner products: $G_{ij} = v_i^\top v_j$.

R 9.2.14 Given a matrix $A \in \mathbb{R}^{m \times n}$, as an abuse of notation, we sometimes also call $AA^\top$ (or $A^\top A$) a Gram matrix of A .

P 9.2.15 Given a real matrix $A \in \mathbb{R}^{m \times n}$, the non-zero eigenvalues of $A^\top A \in \mathbb{R}^{n \times n}$ are the same as the ones of $AA^\top \in \mathbb{R}^{m \times m}$. Both matrices are symmetric and PSD.

P 9.2.16 Every symmetric PSD matrix M is a Gram matrix of an upper triangular matrix C . $M = C^\top C$ is known as the *Cholesky decomposition*.

Singular Value Decomposition

D 9.3.1 (*SVD - Singular Value Decomposition*) Let $A \in \mathbb{R}^{m \times n}$. There exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$, such that $A = U \Sigma V^\top$, where $\Sigma \in \mathbb{R}^{m \times n}$ is a "diagonal" matrix (as in $\Sigma_{ij} = 0$ when $i \neq j$) and the diagonal elements are non-negative and ordered in descending order.

The columns $u_1, \dots, u_m$ of U are called the *left singular vectors* of A and are orthonormal. The columns $v_1, \dots, v_n$ of V are called the *right singular vectors* of A and are orthonormal. The diagonal elements of Σ , $\sigma_i = \Sigma_{ii}$ are called the *singular values* of A and are ordered as $\sigma_1 \geq \dots \geq \sigma_{\min\{m,n\}}$.

R 9.3.2 If A has rank r we can write the SVD in a more compact form: $A = U_r \Sigma_r V_r^\top$, where $U_r \in \mathbb{R}^{m \times r}$ contains the first r left singular vectors, $\Sigma_r \in \mathbb{R}^{r \times r}$ is a diagonal matrix with the first r singular values and $V_r \in \mathbb{R}^{n \times r}$ contains the first r right singular vectors.

Let $A \in \mathbb{R}^{m \times n}$ and $A = U \Sigma V^\top$ be its SVD. Then $AA^\top = U(\Sigma \Sigma^\top)U^\top$. So the left singular vectors of A , the columns of U , are the eigenvectors of $AA^\top$ and the singular values of A are the square-roots of the eigenvalues of $AA^\top$. If $m > n$, A has n singular values and $AA^\top$ has m eigenvalues, but the "missing" ones are 0.

Analogously: $A^\top A = V(\Sigma^\top \Sigma)V^\top$, and so the right singular values of A , the columns of V , are the eigenvectors of $A^\top A$ and the singular values of A are the square-root of the eigenvalues of $A^\top A$. If $n > m$, A has m singular values and $A^\top A$ has n eigenvalues, but the "missing" ones are, again, 0.

T 9.3.3 Every matrix $A \in \mathbb{R}^{m \times n}$ has an SVD decomposition of the form shown above. In other words, every linear transformation is diagonal when viewed in the bases of the singular vectors.

P 9.3.4 Let $A \in \mathbb{R}^{m \times n}$ with rank r . Let $\sigma_1, \dots, \sigma_r$ be the non-zero singular values of A , $u_1, \dots, u_r$ the corresponding left singular vectors and $v_1, \dots, v_r$ the corresponding right singular vectors. Then $A = \sum_{k=1}^r \sigma_k u_k v_k^\top$.

10 Algorithms and Procedures

Gauss Elimination - $O(m^3)$

Gauss elimination is an algorithm that can be used to solve systems of equations of the form $Ax = b$, where $A \in \mathbb{R}^{m \times m}$ and $x, b \in \mathbb{R}^m$.

Back substitution - $O(m^2)$ If we have an upper triangular matrix, we start by scaling the last row, so that the entry becomes one. Then we subtract that row from the second last, scale that one again, subtract both the last and the second last from the third last and continue until we arrive at the top. An example:

$$\begin{aligned} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} x &= \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix} \xrightarrow{(3) \div 7} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 19 \\ 17 \\ 2 \end{bmatrix} \\ \xrightarrow{(2) - 6(3)} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix} x &= \begin{bmatrix} 19 \\ 5 \\ 2 \end{bmatrix} \xrightarrow{(2) \div 5} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 19 \\ 1 \\ 2 \end{bmatrix} \\ \xrightarrow{(1) - 4(3)} \begin{bmatrix} 2 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} x &= \begin{bmatrix} 11 \\ 1 \\ 2 \end{bmatrix} \xrightarrow{(1) - 3(2)} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} x = \begin{bmatrix} 8 \\ 1 \\ 2 \end{bmatrix} \\ \xrightarrow{(1) \div 2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} x &= \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

Elimination If the matrix is not upper triangular, we can transform the system $Ax = b$ into an equivalent system $Ux = c$ with the same solutions, where U is an upper triangular matrix. We do this by starting at the top and, using the diagonal entry of the current column (called the pivot, marked in bold in the example) to eliminate the nonzeros below it with row extractions. An example:

$$\begin{aligned} \begin{bmatrix} 2 & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} x &= \begin{bmatrix} 19 \\ 55 \\ 50 \end{bmatrix} \xrightarrow{(2) - 2(1)} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix} x = \begin{bmatrix} 19 \\ 17 \\ 50 \end{bmatrix} \\ \xrightarrow{(3) - (1)} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 5 & 13 \end{bmatrix} x &= \begin{bmatrix} 19 \\ 17 \\ 31 \end{bmatrix} \xrightarrow{(3) - (2)} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} x = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix} \end{aligned}$$

The *elimination matrix* $E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ means "subtract 2 · (row 1) from (row 2).

Row exchanges When the pivot element is 0, we cannot use it to eliminate nonzeros below it. If there is some nonzero entry below the pivot, we can perform a row exchange to obtain a nonzero pivot.

$$\begin{aligned} \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 8 & 17 \end{bmatrix} x &= \begin{bmatrix} 2 \\ 10 \\ 25 \end{bmatrix} \xrightarrow{(2) \leftrightarrow (1)} \begin{bmatrix} 4 & 6 & 14 \\ 2 & 3 & 4 \\ 2 & 8 & 17 \end{bmatrix} x = \begin{bmatrix} 10 \\ 2 \\ 25 \end{bmatrix} \\ \xrightarrow{(3) - (1)} \begin{bmatrix} 4 & 6 & 14 \\ 0 & 0 & 14 \\ 0 & 5 & 13 \end{bmatrix} x &= \begin{bmatrix} 2 \\ 6 \\ 23 \end{bmatrix} \xrightarrow{(2) \leftrightarrow (3)} \begin{bmatrix} 4 & 6 & 14 \\ 0 & 5 & 13 \\ 0 & 0 & 6 \end{bmatrix} x = \begin{bmatrix} 2 \\ 23 \\ 6 \end{bmatrix} \end{aligned}$$

Success and failure It is possible for Gauss elimination to fail, e.g. when we can no longer move zeros out of the diagonal of our matrix by doing row exchanges. There is a pattern to the success or failure of the algorithm; Let $Ax = b$ be a system of m linear equations in m variables. The following two statements are equivalent (T 3.7):

1. Gauss elimination succeeds.
2. The columns of A are linearly independent.

Gauss-Jordan Elimination - $O(m^2(m+n))$

Gauss-Jordan elimination is an algorithm for solving any system $Ax = b$ or detecting that there is no solution. The algorithm is very similar in spirit to Gauss elimination, but works by transforming the matrix to a matrix in RREF instead of an upper triangular.

Direct Solution A is an $m \times n$ matrix in RREF($j_1, j_2, \dots, j_r$). Then there are two cases. If $c_i \neq 0$ for some row $i > r$, then there is no solution. If $c_i = 0$ for all $i > r$, there is a canonical solution: we define x by:

$$x_j = \begin{cases} c_i & \text{if } j = j_i \text{ for some } i \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{array}{cccccccc} & j_1 & j_2 & & j_3 & & j_4 & \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} & \begin{bmatrix} 0 & 1 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} & = & \begin{bmatrix} b_1 \\ b_2 \\ 0 \\ b_3 \\ b_4 \\ 0 \end{bmatrix} \end{array}$$

Elimination We use row operations to transform any system $Ax = b$ into an equivalent system $Rx = c$, where R is in RREF. We proceed column by column, always starting with a row division (or exchange) and going on from there. The nonzero pivots we find will signal the "downward step" columns $j_1, j_2, \dots$ of the resulting matrix.

$$\begin{aligned} \begin{bmatrix} 2 & 4 & 2 & 2 & -2 \\ 6 & 12 & 6 & 7 & 1 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \xrightarrow{(1) \div 2} \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 6 & 12 & 6 & 7 & 1 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \\ \xrightarrow{(2) - 6(1)} \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 7 \\ 4 & 8 & 2 & 2 & 6 \end{bmatrix} \xrightarrow{(3) - 4(1)} \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix} \end{aligned}$$

We move on to column 2:

$$\begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix}$$

In Gauss elimination, we'd stop. Here, we move on to column 3, where

we can make a row exchange:

$$\begin{aligned} & \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & \mathbf{0} & 1 & 7 \\ 0 & 0 & -2 & -2 & 10 \end{bmatrix} \xrightarrow{(2) \leftrightarrow (3)} \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & -2 & -2 & 10 \\ 0 & 0 & \mathbf{0} & 1 & 7 \end{bmatrix} \\ & \xrightarrow{(2) \div -2} \begin{bmatrix} 1 & 2 & 1 & 1 & -1 \\ 0 & 0 & \mathbf{1} & 1 & -5 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix} \xrightarrow{(1)-(2)} \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & \mathbf{1} & 1 & -5 \\ 0 & 0 & 0 & 1 & 7 \end{bmatrix} \end{aligned}$$

No further elimination is necessary here, so we move to column 4.

$$\begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 1 & -5 \\ 0 & 0 & 0 & \mathbf{1} & 7 \end{bmatrix} \xrightarrow{(2)-(3)} \begin{bmatrix} 1 & 2 & 0 & 0 & 4 \\ 0 & 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & \mathbf{1} & -12 \end{bmatrix}$$

We're not only eliminating below, but also above the pivot (the column needs to be transformed into the correct standard unit vector).

Further downward steps are neither possible nor necessary, so we can stop before even having looked at the last column.

CR decomposition from Gauss-Jordan elimination

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 2 & 5 \end{bmatrix} \xrightarrow{\begin{smallmatrix} (2)-2\cdot(1) \\ (3)-3\cdot(1) \end{smallmatrix}} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 2 & -4 \end{bmatrix} \xrightarrow{(3)-2\cdot(2)} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The resulting matrix R is in RREF(1,3), so columns 1 and 3 are independent in A :

$$C = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}, R = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & -2 \end{bmatrix}$$

(Reduced) Row-Echelon Form

$$\begin{array}{ccccccccc} & j_1 & j_2 & & j_3 & & j_4 & & \\ 1 & \begin{bmatrix} 0 & \mathbf{1} & 0 & 2 & 2 & 0 & 2 & 0 & 2 \end{bmatrix} \\ 2 & \begin{bmatrix} 0 & 0 & \mathbf{1} & 2 & 2 & 0 & 2 & 0 & 2 \end{bmatrix} \\ 3 & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \mathbf{1} & 2 & 0 & 2 \end{bmatrix} \\ 4 & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{1} & 2 \end{bmatrix} \\ 5 & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{0} \end{bmatrix} \\ 6 & \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \mathbf{0} \end{bmatrix} \end{array}$$

D 3.13 Let $R = [r_{ij}]_{i=1}^m \begin{smallmatrix} n \\ j=1 \end{smallmatrix}$. R is in *reduced row echelon form* (RREF) if the following holds: There exist $r \leq m$ column indices $1 \leq j_1 < \dots < j_r \leq n$ such that the following two statements hold:

- For $i = 1, \dots, r$, column j_i of R is the standard unit vector e_i .
- All entries r_{ij} "below the staircase" are 0. This happens if:
 - $i > r$ (entry is below row r).
 - $i \leq r$ and $j < j_i$ (entry is in the part of row i to the left of column j_i).

If we want to describe the shape of R precisely, we say that R is in RREF($j_1, \dots, j_r$).

T 3.19 (Computing *inverses* with Gauss-Jordan elimination). Let $A \in \mathbb{R}^{m \times m}$, and let $(R, j_1, \dots, j_r, M)$ be the output of Gauss-Jordan elimination with input (A, I) . Then A is invertible iff $R = I$, and in this case $A^{-1} = M$.

T 3.21 Let $A \in \mathbb{R}^{m \times n}$. $(R, j_1, \dots, j_r, M)$ is the output of Gauss-Jordan elimination. There exists an invertible $m \times m$ matrix M such that $R = MA$ is in RREF($j_1, \dots, j_r$).

Least Squares Approximation

Suppose we have a linear system $Ax = b$ for which no solution exists. A natural approach is to try and find x for which Ax is as close as possible to b :

$$\min_{\hat{x} \in \mathbb{R}^n} \|A\hat{x} - b\|^2 \quad (0.1)$$

Since the set of possible vectors $y = A\hat{x}$ is exactly $\mathbf{C}(A)$, Ax is precisely the projection of b on $\mathbf{C}(A)$. As we've seen, this can be written as:

$$A^T A \hat{x} = A^T b \quad (0.2)$$

Now, a minimizer of (0.1) is also a solution of (0.2), and A has independent columns, the unique minimizer of (0.1) is given by $\hat{x} = (A^T A)^{-1} A^T b$.

Linear regression is the process of fitting a line through data points and can be accomplished using least squares approximation. Let us consider data points $(t_1, b_1), \dots, (t_m, b_m)$ representing some attribute b over time t . If the relation between t and b is explained by a linear relationship, then it makes sense to search for constants $\alpha_0 \in \mathbb{R}$ and $\alpha_1 \in \mathbb{R}$, such that $b_k \approx \alpha_0 + \alpha_1 t_k$. In particular, it is natural to search for α_0 and α_1 that minimize the sum of squares of the errors:

$$\min_{\alpha_0, \alpha_1} \sum_{k=1}^m (b_k - (\alpha_0 + \alpha_1 t_k))^2, \quad \alpha_0, \alpha_1 \in \mathbb{R} \quad (0.3)$$

If we now define $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$ and $A = \begin{bmatrix} 1 & t_1 \\ \vdots & \vdots \\ 1 & t_m \end{bmatrix}$, we can write (0.3) in matrix-vector notation:

$$\min_{\alpha_0, \alpha_1} \|b - A \begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix}\|^2 \quad (0.4)$$

A has independent columns. Hence, the solution to (0.4) is given by:

$$\begin{pmatrix} \alpha_0 \\ \alpha_1 \end{pmatrix} = (A^T A)^{-1} A^T b = \left[\sum_{k=1}^m t_k \quad \sum_{k=1}^m t_k^2 \right]^{-1} \begin{pmatrix} \sum_{k=1}^m b_k \\ \sum_{k=1}^m t_k b_k \end{pmatrix}$$

If we need to find a polynomial $p(t) = \alpha_0 + \alpha_1 t + \dots + \alpha_k t^k$ which is as close as possible to certain datapoints $(t_1, y_1), \dots, (t_m, y_m)$.

Now, we consider the following definitions:

$$A = \begin{bmatrix} 1 & t_1 & t_1^2 & \dots \\ \vdots & \vdots & \vdots & \dots \\ 1 & t_m & t_m^2 & \dots \end{bmatrix}, x = \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_k \end{pmatrix}, b = \begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix}$$

Our solution can be found by solving the normal equations:

$$A^T A x = A^T b$$

Gram-Schmidt Process

The Gram-Schmidt process can be used to construct an orthonormal basis from a regular one: Given n linearly independent vectors $a_1, \dots, a_n$ that span a subspace S , the Gram-Schmidt process constructs n linearly independent orthogonal vectors $q_1, \dots, q_n$ the following way:

- $q_1 = \frac{a_1}{\|a_1\|}$
- For $k = 2, \dots, n$, set:
 - $q_k = a_k - \sum_{i=1}^{k-1} (a_k^\top q_i) q_i$
 - $q_k = \frac{q_k}{\|q_k\|}$

Change of Basis

Let $A^{m \times n}$ be a matrix representing a linear transformation $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $x \in \mathbb{R}^n \mapsto Ax \in \mathbb{R}^m$, with both input and output written in the canonical bases as $x = \sum_{j=1}^n x_j e_j$ and $Ax = \sum_{i=1}^m (Ax)_i e_i$.

Now, let's say we have a basis for $\mathbb{R}^n$ given by $u_1, \dots, u_n$ and one for $\mathbb{R}^m$ given by $v_1, \dots, v_m$ and we want to understand the linear transformation L written in this basis. Then L takes a vector $x = \sum_{j=1}^n \alpha_j u_j$ and outputs $L(x) = \sum_{i=1}^m \beta_i v_i$.

We might now want to compute the matrix B , that takes $\alpha \mapsto \beta$. In other words, such that $B\alpha = \beta$.

For that, let $U \in \mathbb{R}^{n \times n}$ be the matrix with columns $u_1, \dots, u_n$ and let $V \in \mathbb{R}^{m \times m}$ be the matrix with columns $v_1, \dots, v_m$. Then $x = U\alpha$ and $L(x) = V\beta$, and so $\beta = V^{-1}AU\alpha$. Finally, this means that the matrix B , corresponding to the linear transformation L written in the new bases, is $B = V^{-1}AU$.

11 Tips and Tricks

Formulas for 2×2 Matrices

Determinant

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} := \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \xRightarrow{\quad} A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Roots of polynomials of degree 2

Let $f(x)$ be a polynomial in x : $f(x) = ax^2 + bx + c$.

Then the two roots r_1, r_2 of $f(x)$ can be computed as follows:

$$r_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Singular Value Decomposition

To get the SVD of a matrix $A \in \mathbb{R}^{m \times n}$, we first find the matrices $S_L = AA^T \in \mathbb{R}^{m \times m}$ and $S_R = A^T A \in \mathbb{R}^{n \times n}$.

Both matrices are symmetric, meaning that their eigenvectors are orthogonal to each other. Let $\lambda_1, \dots, \lambda_m$ be the eigenvalues of S_L and $\lambda_1, \dots, \lambda_n$ be the eigenvalues of S_R , both ordered decreasingly. $\lambda_1, \dots, \lambda_{\min(m,n)}$ (i.e. the $\min(m,n)$ greatest eigenvalues) are the same for both matrices, the remaining ones are 0. Both matrices are PSD and their eigenvalues are therefore non-negative.

For $i \in [\min(m,n)]$, we define $\sigma_i = \sqrt{\lambda_i}$. This is the i -th largest singular value of A .

We define $\Sigma \in \mathbb{R}^{m \times n}$ as the rectangular diagonal matrix where $\Sigma_{ii} = \sigma_i$.

We define $U \in \mathbb{R}^{m \times m}$ as the orthogonal matrix where the i -th column is the normalized eigenvector of S_L corresponding to λ_i (these are the left singular vectors).

We define $V \in \mathbb{R}^{n \times n}$ as the orthogonal matrix where the i -th column

is the normalized eigenvector of S_R corresponding to λ_i (there are the right singular vectors).

Now, these three matrices can be combined into the final SVD:

$$A = U\Sigma V^\top$$

SVD of $vw^\top$

Let $n \in \mathbb{N}^+$, Consider arbitrary non-zero vectors $v, w \in \mathbb{R}^n$. $A \in \mathbb{R}^{n \times n}$ (which is a matrix of rank at most 1) has SVD:

$$A = U\Sigma V^\top = \frac{v}{\|v\|}(\|v\| \cdot \|w\|) \left(\frac{w}{\|w\|} \right)^\top$$

Some Useful Equivalences

- Invertibility ($A \in \mathbb{R}^{n \times n}$):
 - A is invertible (A^{-1} exists)
 - $\det(A) \neq 0$ (A is non-singular)
 - 0 is **not** an eigenvalue ($\forall i \lambda_i \neq 0$)
 - $\text{rank}(A) = n$ (full rank)
 - The columns of A are linearly independent
 - $Ax = 0$ has only the trivial solution ($N(A) = \{0\}$)
 - $Ax = b$ has a unique solution for every $b \in \mathbb{R}^n$
 - The RREF of A is I
 - All singular values are non-zero ($\forall i \sigma_i > 0$)
- Linear independence ($v_1, \dots, v_k \in \mathbb{R}^n$):
 - Vectors $v_1, \dots, v_k$ are linearly independent
 - $\sum_{i=1}^k c_i v_i = 0 \implies \forall i c_i = 0$
 - None of the vectors is a linear combination of the others (nor of the preceding ones)
 - The matrix $A = [v_1 \dots v_k]$ has full column rank ($N(A) = \{0\}$)
- Diagonalizability ($A \in \mathbb{R}^{n \times n}$):
 - A is diagonalizable ($A = V\Lambda V^{-1}$)
 - A has n linearly independent eigenvectors
 - The eigenvectors of A form a basis for $\mathbb{R}^n$
 - For every eigenvalue λ : Geom. mult. = Alg. mult.
 - The sum of all Geom. mult. is n

Note: n distinct eigenvalues $\implies$ diagonalizable
Note: A symmetric $\implies$ orthogonally diagonalizable
- Orthogonal matrix ($Q \in \mathbb{R}^{n \times n}$):
 - Q is orthogonal
 - $Q^\top Q = I$
 - $Q^{-1} = Q^\top$
 - The columns of Q form an orthonormal basis of $\mathbb{R}^n$
 - The rows of Q form an orthonormal basis of $\mathbb{R}^n$
 - $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$ and $(Qx) \cdot (Qy) = x \cdot y$ for all $x, y \in \mathbb{R}^{n \times n}$
- Positive (semi-)definite matrix ($A = A^\top \in \mathbb{R}^{n \times n}$)
 - $\lambda_i > 0$ ($\geq$) for $i = 1, \dots, n$
 - $x^\top A x > 0$ ($\geq$) for all $x \neq 0$
 - All pivots (from gaussian elimination without swaps) are > 0
 - All upper-left sub-determinants are > 0 (Sylvester's Criterion)

Quick Projections

- Projection p of b onto a :

$$p = \frac{a^\top b}{a^\top a} a$$

Check: $e = b - p$ orthogonal to a (i.e., $a^\top e = 0$)

- Find $\hat{x}$ that minimizes $\|Ax - b\|^2$ (least squares approx.):
 - Calculate $M = A^\top A$
 - Calculate $d = A^\top b$
 - Solve $M\hat{x} = d$

Note: if A has independent columns, $\hat{x} = (A^\top A)^{-1} A^\top b$

- Projection of b onto $S \subseteq \mathbb{R}^m$
 - Find basis for S , construct matrix A
 - Find $\hat{x}$ using least squares approx.
 - Find $p = A\hat{x}$

Note: Projection matrix: $P = A(A^\top A)^{-1} A^\top$

Diagonalization ($A = V\Lambda V^{-1} \in \mathbb{R}^{n \times n}$)

- Find eigenvalues $\lambda_1, \dots, \lambda_n$
- Find n independent eigenvectors v_i (if there aren't n , A is not diagonalizable)
- Construct matrices:
 - $\Lambda \in \mathbb{R}^{n \times n}$ has λ_i on the diagonal
 - $V \in \mathbb{R}^{n \times n}$ has v_i as i -th column

The **Spectral Decomposition** is basically the diagonalization of a symmetric matrix ($A = A^\top$):

- All eigenvalues will be real
- As there are n eigenvalues, there are n independent v_i
- Eigenvectors are normalized ($\|v_i\| = 1$)
- We write $A = Q\Lambda Q^\top$ (Where Q is orthogonal)

Quick Sanity-Checks

- Trace:** $\text{Tr}(A) = \sum a_{ii} = \sum \lambda_i$ (trace = sum of EW)
- Determinant:** $\det(A) = \prod \lambda_i$ (det = prod of EW)
- Rank:** $\text{rank}(A) = \dim(\mathbf{C}(A)) = \dim(\mathbf{R}(A)) = \# \text{ non-zero SW}$
- Orthogonality:** $Q^\top Q = I$, $\det(Q) = \pm 1$, $\|Qx\| = \|x\|$

Common Proofs

- Prove that $U \subseteq V$ is a subspace:
 - Show $0 \in U$
 - Let $u, v \in U$, $\lambda \in \mathbb{R}$. Show $\lambda u + v \in U$.

Alternatively, a good counterexample is usually showing closure doesn't hold for some vector, or $0 \notin U$
- Prove $v_1, \dots, v_k \in \mathbb{R}^n$ linearly independent:
 - Set up equation $\sum \lambda_i v_i = 0$
 - Show how this implies $\lambda_1 = \dots = \lambda_k = 0$

Alternatively, use the vectors as columns of $A \in \mathbb{R}^{n \times k}$ and show $\mathbf{N}(A) = \{0\}$ (or $\text{rank}(A) = k$)
- Prove properties of $T: V \rightarrow W$:
 - Injectivity** (1-to-1): $\ker(T) = \{0\}$ ($Ax = 0 \implies x = 0$)
 - Surjectivity** (onto): $\text{Im}(T) = W$ ($\text{rank}(A) = \dim(W)$)
 - Bijectivity:** Show both (or just one if $\dim(V) = \dim(W)$)
- Prove properties of matrix A :
 - Symmetry:** Show $A^\top = A$ (reminder: $(AB)^\top = B^\top A^\top$)
 - Orthogonality:** Show $Q^\top Q = I$
 - PD or PSD:** Show $x^\top A x > 0$ or $x^\top A x \geq 0$ for all $x \neq 0$

Random Matrix Algebra Hacks

- Matrices constructed with the standard basis: Let $E_{ij} = e_i e_j^\top$ (Matrix with only $e_{ij} = 1$). Then $E_{ij} E_{kl} = (e_i e_j^\top)(e_k e_l^\top) = e_i (e_j^\top e_k) e_l^\top$. This means that $E_{ij} E_{kl} = 0$ unless $j = k$
- Matrices in the form $M = \begin{bmatrix} A & B \\ 0 & D \end{bmatrix}$:
 - $\det(M) = \det(A) \cdot \det(D)$.
 - $\{\text{EW of } M\} = \{\text{EW of } A\} \cup \{\text{EW of } D\}$
- Rank properties
 - $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$
 - $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
 - $\text{rank}(A) + \text{rank}(B) \leq \text{rank}(AB) + n$ (where $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times k}$)
 - $\text{rank}(A) = \dim(\mathbf{C}(A)) = \dim(\mathbf{R}(A))$
 - $\text{rank}(A) + \dim(\mathbf{N}(A)) = n$